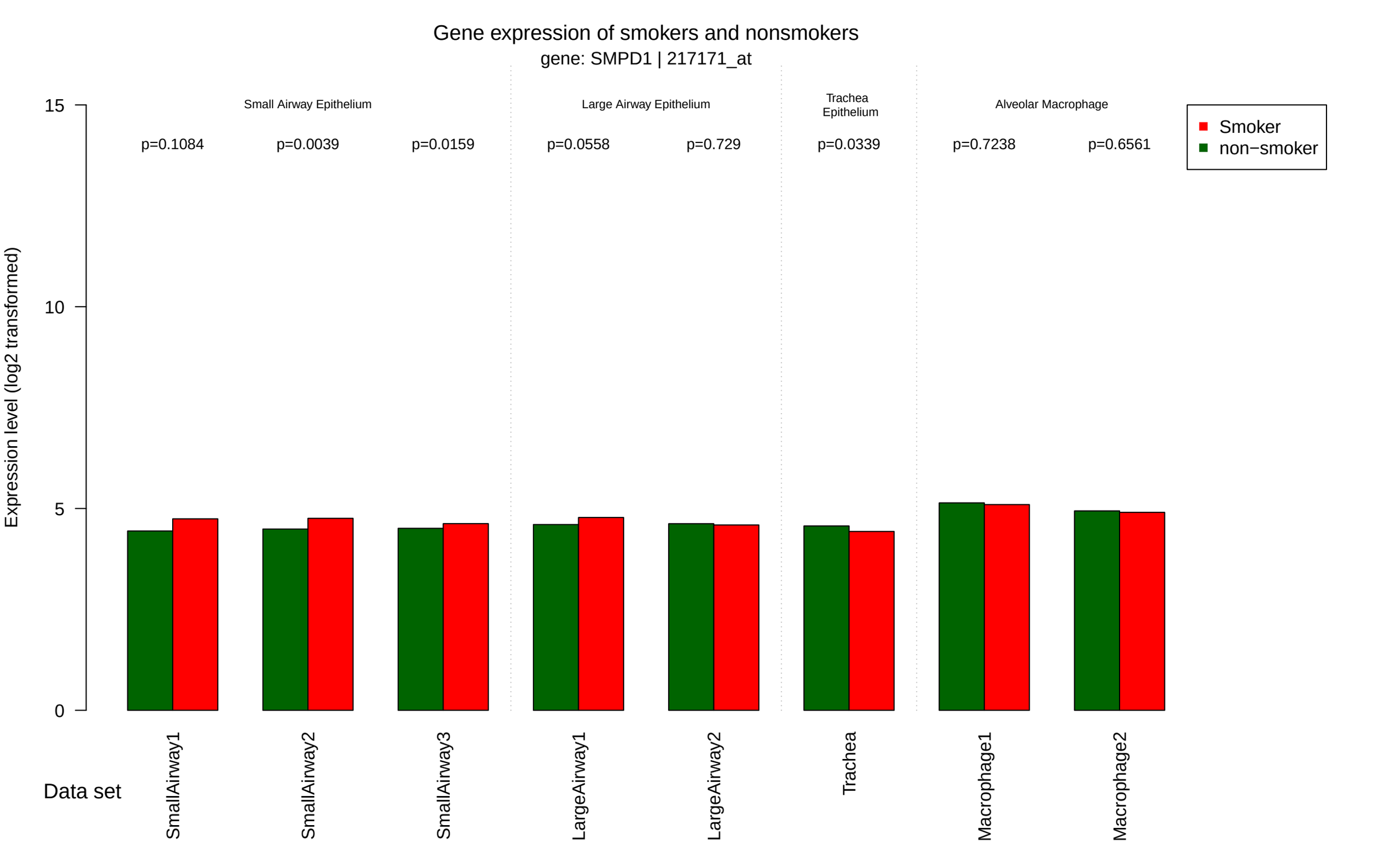
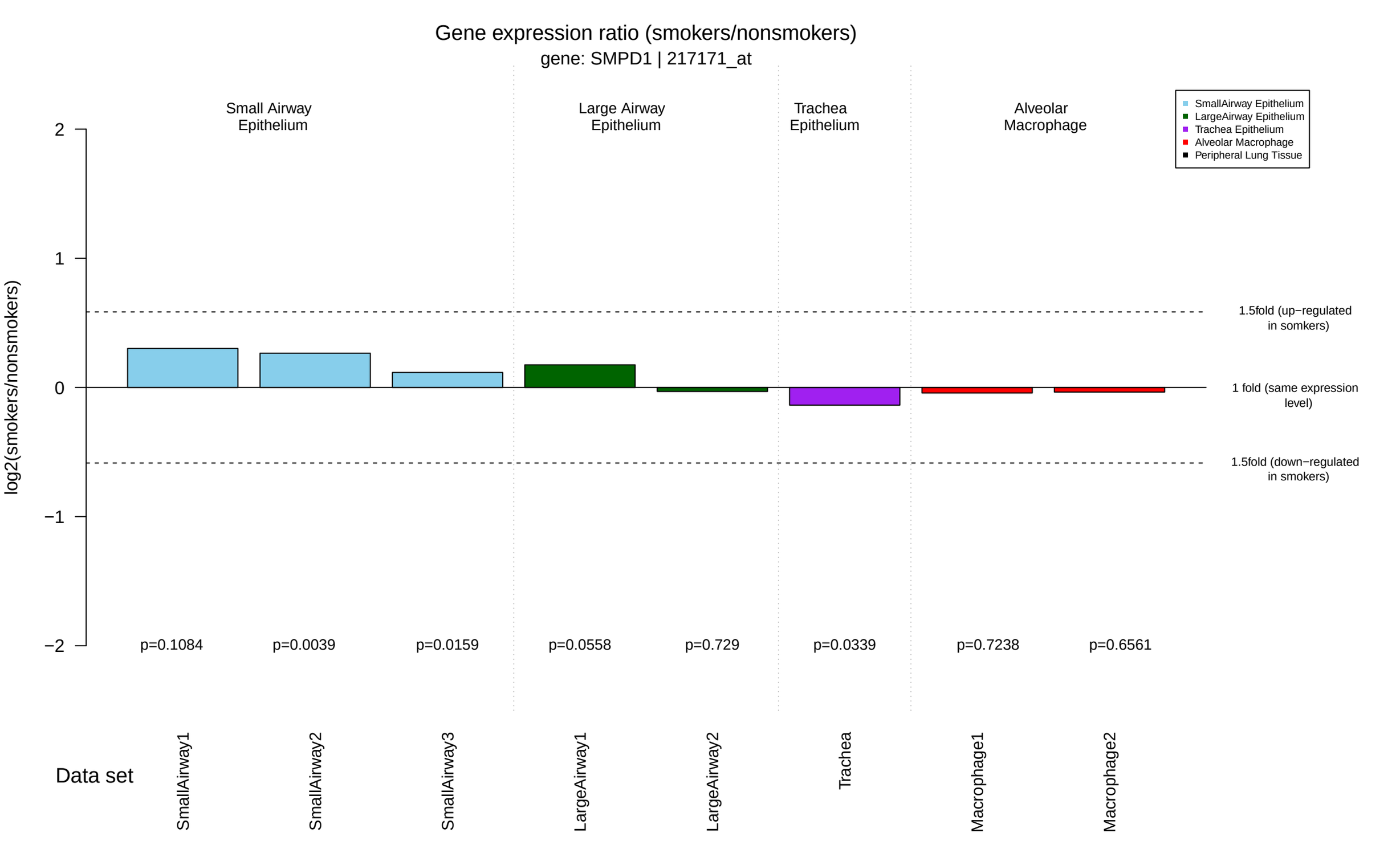
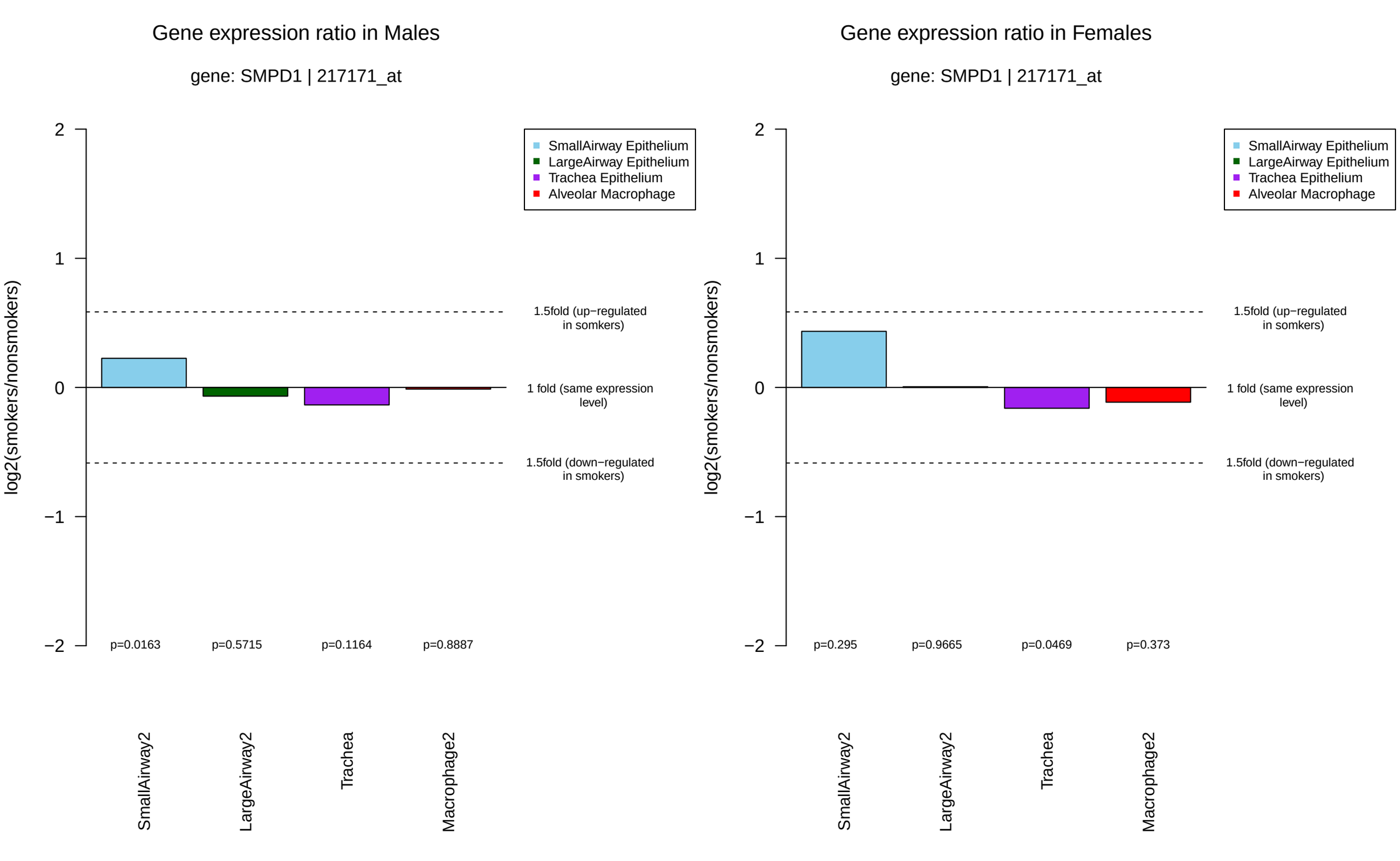
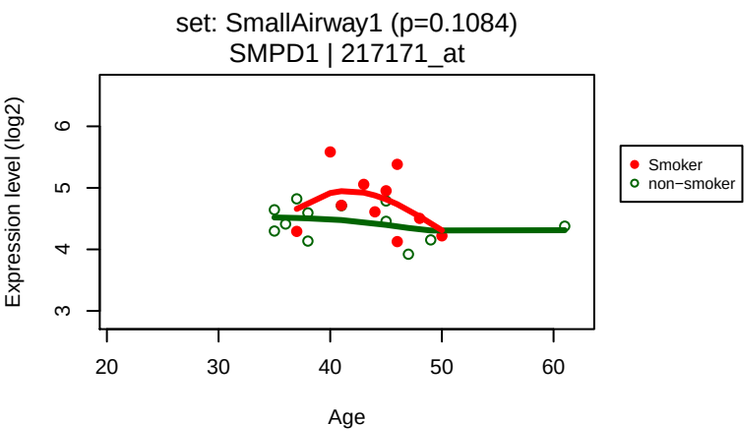
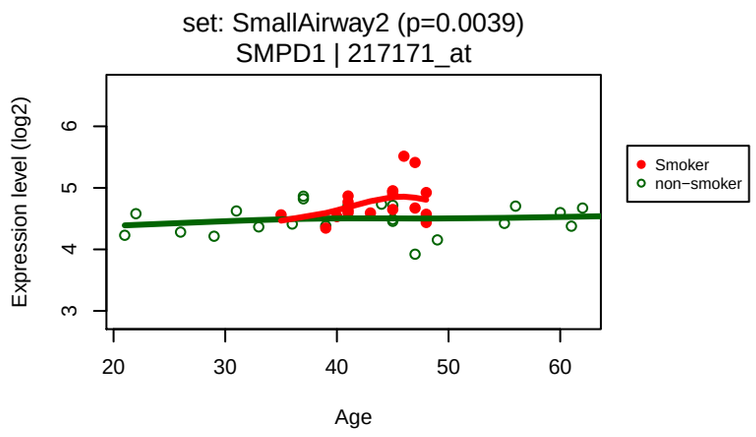
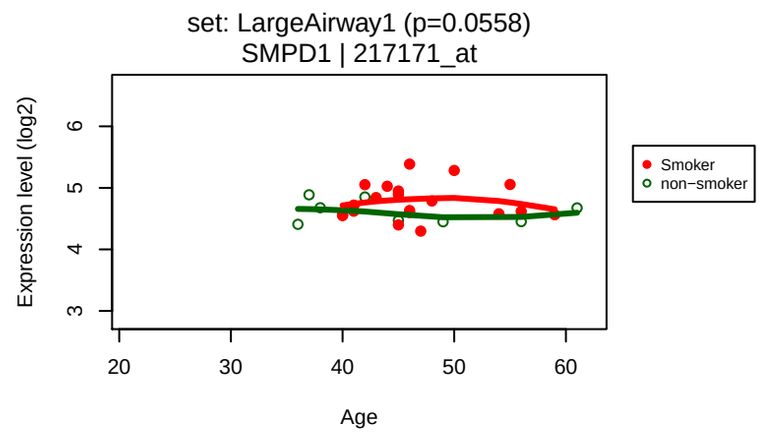

Gene expression of smokers and nonsmokers
gene: SMPD1 | 217171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1084
p=0.0039
p=0.0159
p=0.0558
p=0.729
p=0.0339
p=0.7238
p=0.6561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

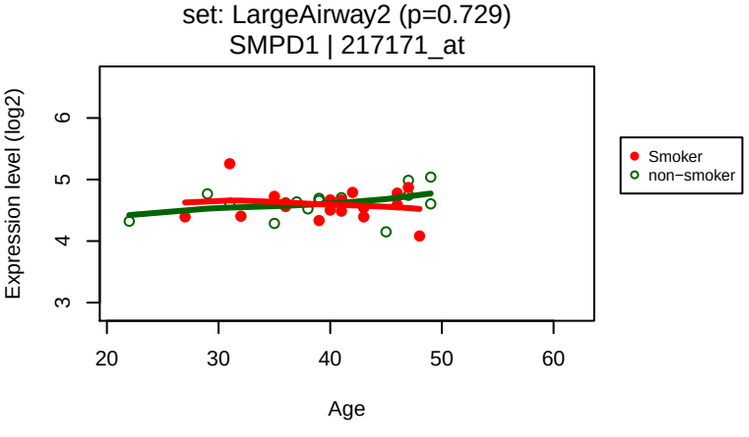

Gene expression ratio (smokers/nonsmokers)
gene: SMPD1 | 217171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1084
p=0.0039
p=0.0159
p=0.0558
p=0.729
p=0.0339
p=0.7238
p=0.6561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

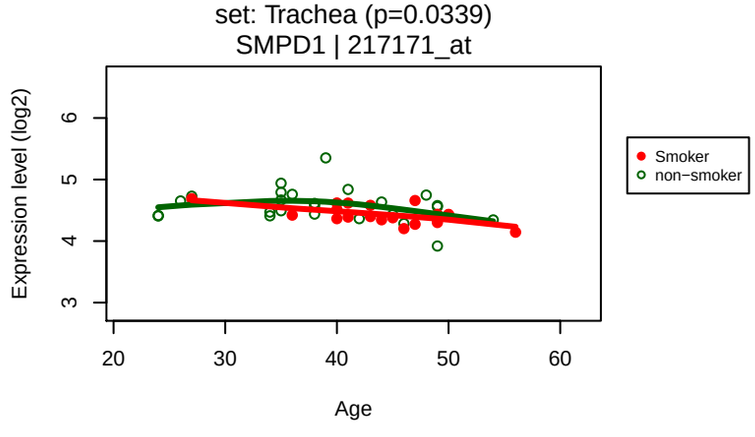

Gene expression ratio in Males
gene: SMPD1 | 217171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0163
p=0.5715
p=0.1164
p=0.8887
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMPD1 | 217171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.295
p=0.9665
p=0.0469
p=0.373
SmallAirway2
LargeAirway2
Trachea
Macrophage2

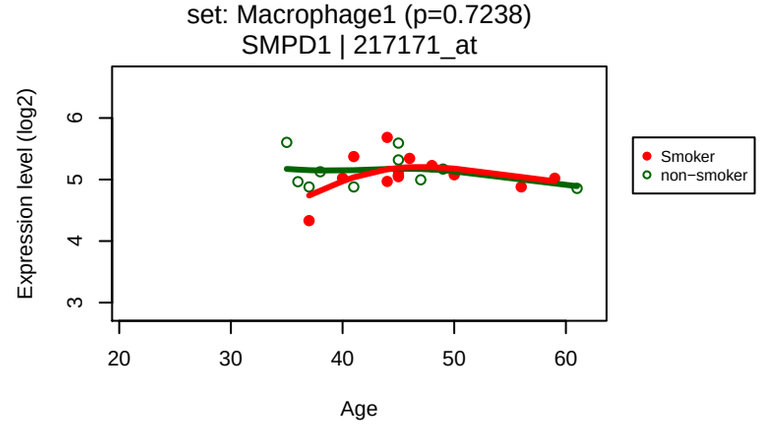

set: SmallAirway1 (p=0.1084)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker

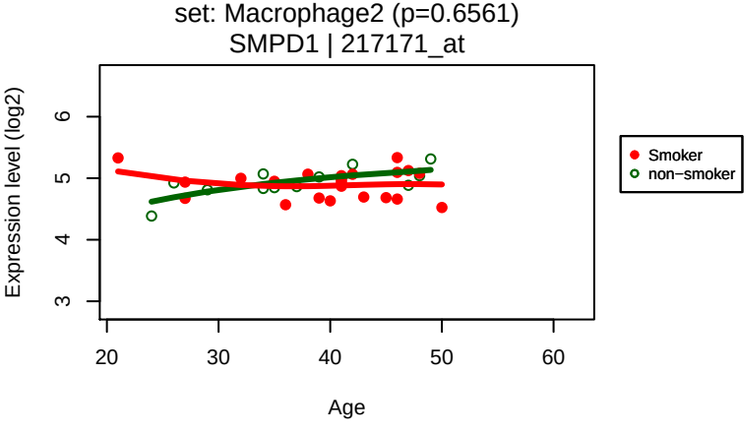

set: SmallAirway2 (p=0.0039)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker

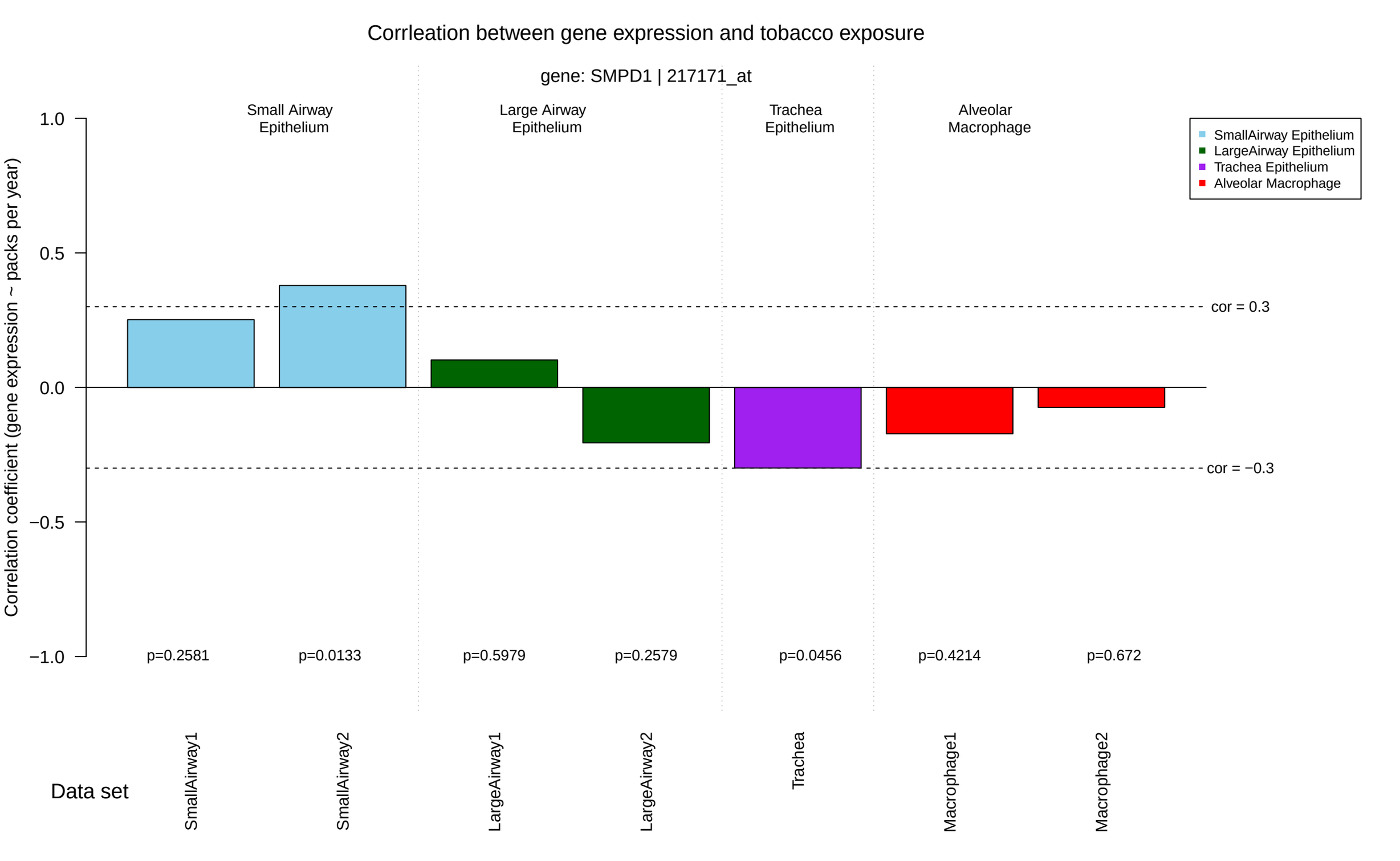

set: LargeAirway1 (p=0.0558)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.729)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0339)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7238)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6561)
SMPD1 | 217171_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMPD1 | 217171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2581
p=0.0133
p=0.5979
p=0.2579
p=0.0456
p=0.4214
p=0.672
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2